PARKINSON'S PROGRESSION MARKERS INITIATIVE

Play a Part in Parkinson's Research

Measurement of PPMI Samples with Elecsys Neuro Toolkit Immunoassays

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PPMI Project ID: 152

Summary

Two (2) sets of 1660 pristine aliquots were analyzed by electrochemiluminescence immunoassays (ECLIA) on the fully automated **cobas e** 411 with eight (8) Roche Elecsys immunoassays. The testing was performed over two (2) separate testing periods (Phase 1 and Phase 2).

	Phase 1	Phase 2
Testing Period	May 23 through June 12, 2019	Oct 8 through Nov 14, 2019
# of patient samples	385	1231
# of reference pools	10	34
Total # of samples	395	1265

The specific lots used for each of the 8 biomarkers in each testing period are outlined in Table 1 below.

Table 1: Table showing the names of each biomarker tested with the respective Elecsys Immunoassay and the lot numbers used

					QC Acceptance Criteria (Kit Lot Specific)			QC Acceptance Criteria (Kit Lot Specific)	
Abbreviated Name	Unabbreviated Name	Unit of Measure	Measuring Range	Lot # Phase 1 (n=395)	Lower Control Range	Upper Control Range	Lot # Phase 2 (n=1265)	Lower Control Range	Upper Control Range
α-syn	Alpha synuclein	pg/mL	1.5 - 3375	**K02-0418**	167 - 256	653 - 1001	**K02-0418**	167 - 256	653 - 1001
GFAP[1]	Glial fibrillary acid protein	ng/mL	0.004 - 198	**R01-0818**	2.4 – 3.7	44.7 – 68.4	**K02-0519**	2.45 – 3.76	46.1 – 70.6
IL-6	Interleukin 6	pg/mL	1.5 - 5000	**K02-0119**	30.18 – 46.22	193.6 – 296.5	**K02-0119**	30.18 – 46.22	193.6 – 296.5
S100b[1]	S-100 calcium-binding protein B	ng/mL	0.015 - 30	**31100201**	0.158 – 0.242	1.92 – 2.94	**K02-0719**	0.166 – 0.254	2.09 – 3.19
NG[2]	Neurogranin	pg/mL	5.788 - 5000	**K01-1018**	68.88 – 105.50	1447.36 – 2216.84	**K01-1018**	68.88 – 105.50	1447.36 – 2216.84
NF-L	Neurofilament Light	pg/mL	0.21 - 1997	**K01-1218**	11.5 – 17.6	376 - 576	**K01-1218**	11.5 – 17.6	376 - 576
sTREM2	Soluble triggering receptor expressed on myeloid cells 2	ng/mL	0.0005 - 39.1	**K03-0119**	2.1 – 3.3	27.1 – 41.5	**K03-0119**	2.1 – 3.3	27.1 – 41.5
YKL40[1]	Chitinase-3-like Protein 1	ng/mL	0.04 - 973	**K03-1218**	23.2 – 35.5	112.9 – 173.0	**K04-1218**	23.2 – 35.5	107.5 – 164.7

[1]Denotes assays that had a lot change between phase 1 and phase 2 measurements. Summary data for the lot to lot comparison can be found under "Additional Notes" below.
[2]Denotes assay was tested, but results for phase 2 measurements are pending additional investigation prior to release decision.

All eight (8) immunoassays are for investigational use only. They are currently under development by Roche Diagnostics and are not commercially available products.

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Method

The 1660 PPMI aliquots were tested with the Roche Elecsys CSF immunoassays at Covance Greenfield Laboratories (Translation Biomarker Solutions), according to the preliminary kit manufacturer's instructions and as described by Roche's Exploratory Study Protocol, "Measurements of Samples with the Elecsys Neuro Toolkit Immunoassays". Fifty-eight (58) PD and CT PPMI reference pools were also included in the 1660 subset of PPMI samples. Twenty (20) reference pools were run during the Phase 1 testing period while thirty-eight (38) were run during the Phase 2 testing period. All sample results and reference pool results will be made available.

Samples were thawed for 30 minutes at room temperature in an upright position. After thawing, roller mixer was used to mix the samples for 20 minutes, avoiding foam formation. Elecsys measurements were performed in a series of runs. Each sample was run in singlicate for each of the 8 assays, over the two time periods as shown in table 2 and 3.

Table 2: Table showing aliquot usage, testing days and number of samples run in each batch for testing period #1: Phase 1 (May 23 through June 12, 2019)

Aliquot #1	Testing Days				
	May 23	May 31	June 3	June 4	June 5
α-syn	80	80	80	80	75
GFAP	80	80	80	80	75
IL-6	80	80	80	80	75
S100b	80	80	80	80	75
Aliquot #2	Testing Days				
	June 6	June 7	June 10	June 11	June 12
NG	80	80	80	75	80
NF-L	80	80	80	75	80
sTREM2	80	80	80	75	80
YKL40	80	80	80	75	80

*Table 3: Table showing aliquot usage, testing days and number of samples run in each batch for testing time period #2: Phase 2 (**Oct 8 through Nov 14, 2019)***

Aliquot #2	Assay			
	NG	NF-L	sTREM2	YKL40
Oct 8	80			
Oct 9	30			
Oct 10	78 + 78			
Oct 11	78 + 80			
Oct 14	80			
Oct 15	80			
Oct 16	80 + 78			
Oct 18	78			
Oct 22	79			
Oct 24	78 + 79			
Oct 25	79			
Oct 28	79 + 51			
Aliquot #1	Assay			
	α-syn	GFAP	IL-6	S100b
Nov 1	80			
Nov 4	30 + 78			
Nov 5	78 + 78			
Nov 6	80 + 80			
Nov 7	80 + 80			
Nov 11	78 + 78			
Nov 12	79 + 78			
Nov 13	79 + 79			
Nov 14	79 + 51			

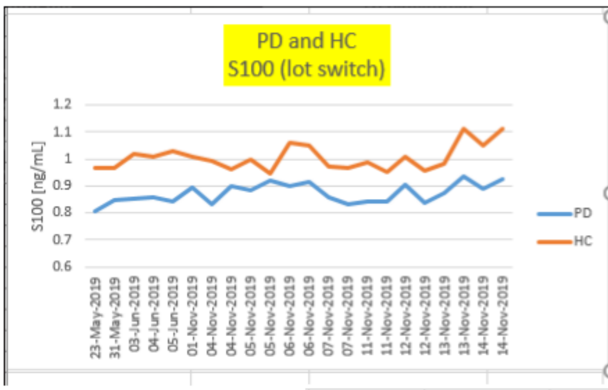
Beginning each of the 31 testing days, calibration and quality controls were run. Where two (2) batches of samples were measured in a single day, the second rack pack was also calibrated and controls were run. Results were within stated limits to meet acceptance criteria. The measuring ranges and units as well as the QC Acceptance Criteria applied for each immunoassay are shown in table 1 above.

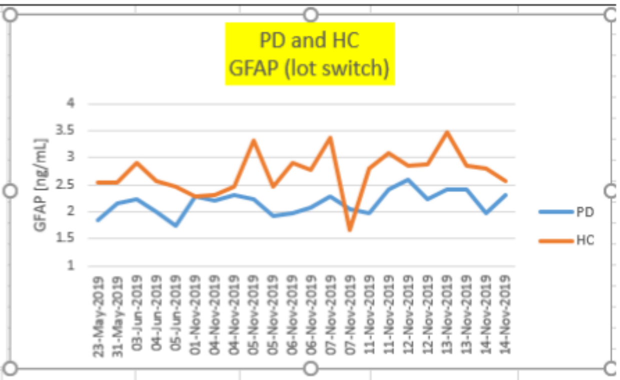
Additional Notes

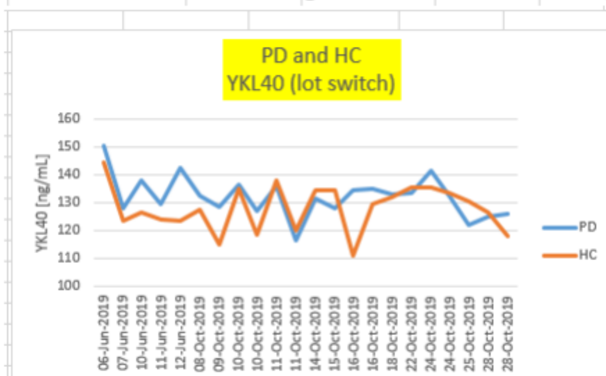
Data in this section supports the lot-to-lot consistency between the lots for those assays experiencing a lot change between testing phases. The data shows that there should be no significant impact on the results and any subsequent analyses attributed to the lot change for these biomarkers.

PPMI Reference Pool Data:

Graphs of PPMI Reference Pools (PD and HC) over both testing periods (for assays where a lot changed occurred between the phase 1 and phase 2 testing periods):

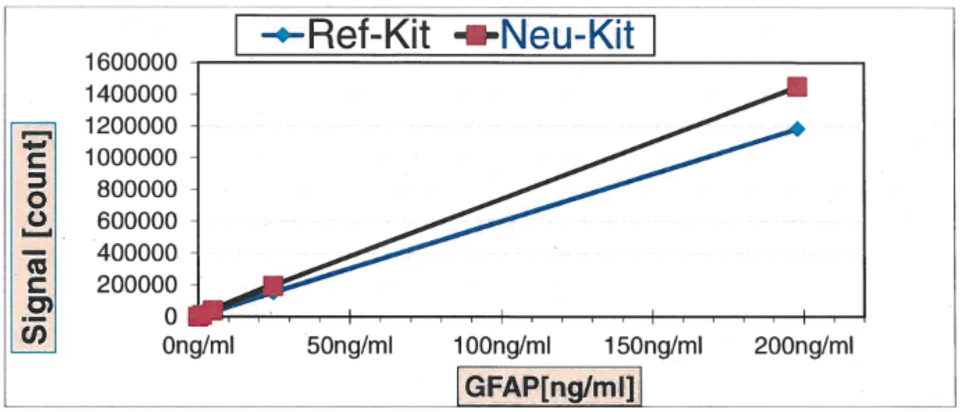




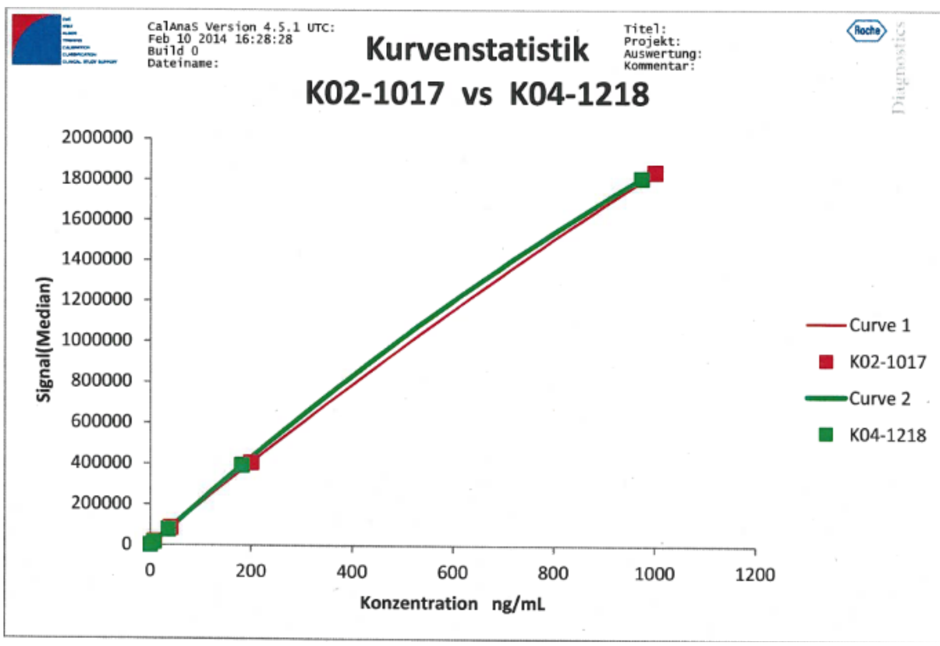


Roche (Early Stage Development) R&D Lot Rollover Summary:

Assay	# of Samples in MC	Slope	Correlation	Signal of new vs prior
S100b	CE launched IVD product			
GFAP	10	1.04	1.00	122%
YKL40	10	0.96	0.99	114%



GFAP Method Comparison (Roche R&D data)



YKL40 Method Comparison (Roche R&D data)

Note: K02-1017 is the same lot as K03-1018 (Kit K03-1018 consists of Reagent, R02-1017, Calibrator, C03-1218 and Control, Q03-1218)

References

1. Molinuevo JL, Ayton S, Batrla R, Bednar MM, Bittner T, Cummings J, Fagan AM, Hampel H, Mielke MM, Mikulskis A, O'Bryant S, Scheltens P, Sevigny J, Shaw LM, Soares HD, Tong G, Trojanowski JQ, Zetterberg H, Blennow K. Current state of Alzheimer's fluid biomarkers. Acta Neuropathol. 2018 Dec;136(6):821-853.

About the Authors

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